MATRICES

Introduction

The knowledge of matrices is necessary in various branches of mathematics. Matrices are one of the most powerful tools in mathematics. This mathematical tool simplifies our work to a great extent when compared with other straight forward methods. The evolution of concept of matrices is the result of an attempt to obtain compact and simple methods of solving system of linear equations. Matrices are not only used as a representation of the coefficients in system of linear equations, but utility of matrices far exceeds that use. Matrix notation and operations are used in electronic spreadsheet programs for personal computer, which in turn is used in different areas of business and science like budgeting, sales projection, cost estimation, analysing the results of an experiment etc. Also, many physical operations such as magnification, rotation and reflection through a plane can be represented mathematically by matrices. Matrices are also used in cryptography. This mathematical tool is not only used in certain branches of sciences, but also in genetics, economics, sociology, modern psychology and industrial management.

Suppose we wish to express the information that Radha has 15 notebooks. We may express it as [15] with the understanding that the number inside [ ] is the number of notebooks that Radha has. Now, if we have to express that Radha has 15 notebooks and 6 pens. We may express it as [15 6] with the understanding that first number inside [ ] is the number of notebooks while the other one is the number of pens possessed by Radha. Let us now suppose that we wish to express the information of possession of notebooks and pens by Radha and her two friends Fauzia and Simran which is as follows:

Radha	has	15	notebooks	and	6	pens,
Fauzia	has	10	notebooks	and	2	pens,
Simran	has	13	notebooks	and	5	pens

Now this could be arranged in the tabular form as follows:

	Notebook	**Pens**
Radha	15	6
Fauzia	10	2
Simran	13	5

and this can be expressed as

$$\underbrace{\begin{bmatrix} 15 & 6 \\ 10 & 2 \\ 13 & 5 \end{bmatrix}}_{\text{2 coulmns}} \left.\vphantom{\begin{bmatrix} 15 \\ 10 \\ 13 \end{bmatrix}}\right\} \text{3 rows}$$

or

	Radha	Fauzia	Simran
Notebooks	15	10	13
Pens	6	2	5

which can be expressed as:

$$\underbrace{\begin{bmatrix} 15 & 10 & 13 \\ & & \\ 6 & 2 & 5 \end{bmatrix}}_{\text{3 coulmns}} \left.\vphantom{\begin{bmatrix} 1 \\ \\ 1 \end{bmatrix}}\right\} \text{2 rows}$$

In the first arrangement the entries in the first column represent the number of note books possessed by Radha, Fauzia and Simran, respectively and the entries in the second column represent the number of pens possessed by Radha, Fauzia and Simran, respectively. Similarly, in the second arrangement, the entries in the first row represent the number of notebooks possessed by Radha, Fauzia and Simran, respectively. The entries in the second row represent the number of pens possessed by Radha, Fauzia and Simran, respectively. An arrangement or display of the above kind is called a matrix. Formally, we define matrix as:

Definition 0.1. *A $m \times n$ matrix A is a rectangular array of m n real (or complex numbers) arranged in m horizontal rows an n vertical columns.*

(i) We shall follow the notation, namely $A = [a_{ij}]_{m\times n}$ to indicate that A is a matrix of order $m \times n$.

(ii) We shall consider only those matrices whose elements are real numbers or functions taking real values.

The **dimension or order** of a matrix A is determined by the number of rows and columns of the matrix. If a matrix A has m rows and n columns we denote its dimension or order by $m \times n$ and read as "m by n".

For example, A = $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ is a 2×2 matrix and B = $\begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 3 \end{bmatrix}$ is a 2×3 order matrix.

Note: Note a that an $m \times n$ matrix has mn elements.

Type of Matrices

In this section, we shall discuss different types of matrices

Square Matrix

Definition 0.2. *A square matrix is one in which the number of rows is equal to the number of columns.*

For instance, A= $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, B= $\begin{bmatrix} 3 & -1 & 6 \\ 8 & 2 & 9 \\ 0 & 7 & 3 \end{bmatrix}$, C= $\begin{bmatrix} 5 & -1 & 8 & 3 \\ 2 & 5 & 7 & 8 \\ 3 & 6 & 11 & 0 \\ 0 & -1 & 8 & 7 \end{bmatrix}$, are square Matrices.

Note: If a square matrix has n rows (and thus n columns), then A is said to be a square matrix of order n.

Diagonal Matrix

Definition 0.3. *A square matrix* $A[a_{ij}]_{n\times n}$ *for which* $a_{ij} = 0$ *for* $i \neq 0$*, is called diagonal matrix.*

For instance, D= $\begin{bmatrix} 4 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 6 \end{bmatrix}$ and E= $\begin{bmatrix} 10 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 5 \end{bmatrix}$ are diagonal Matrices.

Scalar Matrix

Definition 0.4. *A diagonal matrix* $A = [a_{ij}]_{n\times n}$ *for which all the terms on the main diagonal are equal, that is* $a_{ij} = k$ *for* $i = j$ *and* $a_{ij} = 0$ *for* $i \neq j$ *is called a scalar matrix.*

For instance, H= $\begin{bmatrix} -5 & 0 & 0 \\ 0 & -5 & 0 \\ 0 & 0 & -5 \end{bmatrix}$ and I= $\begin{bmatrix} 4 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix}$ are scalar Matrices.

Unit or Identity Matrix

Definition 0.5. *A square matrix* $A = [a_{ij}]_{n\times n}$ *is said to be the unit matrix or identity matrix if*

$$a_{ij} = \begin{cases} 0 & \text{if} \quad i \neq j \\ 1 & \text{if} \quad i = j \end{cases}$$

For example, $I_1 = [1]$, I_2= $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ and I_3= $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

Row Matrix or Column Matrix

Definition 0.6. *A matrix with just one row of elements is called a row matrix or row vector. While a matrix with just one column of elements is called a column matrix or column vector.*

For instance, A = [2 5 -15] is a row matrix whereas B= $\begin{bmatrix} 3 \\ 7 \\ 5 \end{bmatrix}$ is a column matrix.

Zero matrix or Null matrix

Definition 0.7. *An $m \times n$ matrix is called a zero matrix or null matrix if each of its elements is zero.*

For example, $Z_1 = [0]$, $Z_2 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ and $Z_3 = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

Equal matrix

Two matrices $A = [a_{ij}]$ and $B = [b_{ij}]$ are said to be equal if

(i) they are of same order.

(ii) each element of A is equal to the corresponding element of B, that is $a_{ij} = b_{ij}$ for all i and j.

Example 1. *$A = \begin{bmatrix} 2 & 3 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 3 \\ 0 & 1 \end{bmatrix}$ are two matrix then A & B are equal matrix. But, if $A = \begin{bmatrix} 3 & 2 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 3 \\ 0 & 1 \end{bmatrix}$ then A & B are not equal matrix. Symbolically, if two matrices A and B are equal, we write A = B.*

Example 2. *If $\begin{bmatrix} x & y \\ z & a \\ b & c \end{bmatrix} = \begin{bmatrix} -1.5 & 0 \\ 2 & \sqrt{6} \\ 3 & 2 \end{bmatrix}$, then the value of x, y, z, a, b, c.*

Solution: *The value of x = -1.5, y = 0, z = 2, a = $\sqrt{6}$, b = 3, c = 2*

Example 3. *If $\begin{bmatrix} x+3 & z+4 & 2y-7 \\ -6 & a-1 & 0 \\ b-3 & -21 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 6 & 3y-2 \\ -6 & -3 & 2c+2 \\ 2b+4 & -21 & 0 \end{bmatrix}$. Find the values of a, b, c, x, y and z.*

Solution: *As the given matrices are equal, therefore, their corresponding elements must be equal. Comparing the corresponding elements, we get*

x + 3 = 0 $\implies$ x=-3, z + 4 = 6 $\implies$ z=2

2y - 7 = 3y - 2 $\implies$ 3y-2y=-7+2 $\implies$ y =-5,

a-1=-3 $\implies$ a=-2

0 = 2c + 2 $\implies$ 2c=-2 $\implies$ c =-1,

b - 3 = 2b + 4 $\implies$ -3-4=2b-b $\implies$ b=-7.

Example 4. *Find the values of a, b, c, and d from the following equation:*

$$\begin{bmatrix} 2a+b & a-2b \\ 5c-d & 4c+3d \end{bmatrix} = \begin{bmatrix} 4 & -3 \\ 11 & 24 \end{bmatrix}$$

Solution: *By equality of two matrices, equating the corresponding elements, we get*

2a + b = 4, 5c - d = 11

a - 2b = - 3 4c + 3d = 24

Solving these equations, we get

a = 1, b = 2, c = 3 and d = 4

Operations on Matrices

In this section, we shall introduce certain operations on matrices, namely, addition of matrices, multiplication of a matrix by a scalar, difference and multiplication of matrices.

Addition of matrices

Let $A = [a_{ij}]$ and $B = [b_{ij}]$ be are two $m \times n$ matrices. Then the sum A + B is defined to be the matrix $C = [c_{ij}]$ with $c_{ij} = a_{ij} + b_{ij}$.

In general, If A = $\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}$ is a 2 × 3 matrix and B= $\begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \end{bmatrix}$ is another 2×3 matrix. Then, we define A+B= $\begin{bmatrix} a_{11}+b_{11} & a_{12}+b_{12} & a_{13}+b_{13} \\ a_{21}+b_{21} & a_{22}+b_{22} & a_{23}+b_{23} \end{bmatrix}$

Example 5. *Given* $A = \begin{bmatrix} \sqrt{3} & 1 & -1 \\ 2 & 3 & 0 \end{bmatrix}$ *and* $B = \begin{bmatrix} 2 & \sqrt{5} & 1 \\ -2 & 3 & \frac{1}{2} \end{bmatrix}$ *, find* $A + B$

Solution: Since A, B are of the same order 2×3. Therefore, addition of A and B is defined and is given by

$$A+B = \begin{bmatrix} \sqrt{3}+2 & 1+\sqrt{5} & -1+1 \\ 2-2 & 3+3 & 0+\frac{1}{2} \end{bmatrix} = \begin{bmatrix} \sqrt{3}+2 & 1+\sqrt{5} & 0 \\ 0 & 6 & \frac{1}{2} \end{bmatrix}$$

Note: The addition of two matrix A and B are possible only if the order of both the matrix are same. If A and B are not of same order, then A + B is not defined.

Scalar Multiplication of a Matrix

Let $A = [a_{ij}]$ be an $m \times n$ matrix. Then for any element $k \in \mathbb{R}$, we define $kA = [ka_{ij}]$, i.e., kA is obtained by multiplying each element of A by the scalar k.

Example 6. *If* $A = \begin{bmatrix} 3 & 1 & 1.5 \\ \sqrt{5} & 7 & -3 \\ 2 & 0 & 5 \end{bmatrix}$

, then

$$3A = 3\begin{bmatrix} 3 & 1 & 1.5 \\ \sqrt{5} & 7 & -3 \\ 2 & 0 & 5 \end{bmatrix} = \begin{bmatrix} 3\times 3 & 3\times 1 & 3\times 1.5 \\ 3\times\sqrt{5} & 3\times 7 & 3\times -3 \\ 3\times 2 & 3\times 0 & 3\times 5 \end{bmatrix} = \begin{bmatrix} 9 & 3 & 4.5 \\ 3\sqrt{5} & 21 & -9 \\ 6 & 0 & 15 \end{bmatrix}$$

Negative of a matrix

The negative of a given matrix A is denoted by $-A$. We define $-A = (-1)A$.

Example 7. *Let* $A = \begin{bmatrix} 3 & 1 \\ -5 & x \end{bmatrix}$. *Then* $-A$ *given by*

$$-A = (-1)A = (-1)\begin{bmatrix} 3 & 1 \\ -5 & x \end{bmatrix} = \begin{bmatrix} -3 & -1 \\ 5 & -x \end{bmatrix}$$

Difference of matrices

If $A = [a_{ij}]$, $B = [b_{ij}]$ are two matrices of the same order, say $m \times n$, then difference $A - B$ is defined as a matrix $D = [d_{ij}]$, where $d_{ij} = a_{ij} - b_{ij}$, for all value of i and j. In other words, $D = A - B = A + (-1)B$, that is sum of the matrix A and the matrix $-B$.

Example 8. *If* $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix}$ *and* $B = \begin{bmatrix} 3 & -1 & 3 \\ -1 & 0 & 2 \end{bmatrix}$, *then find* $2A - B$.

Solution: We have

$$\begin{aligned} 2A - B = 2A + (-1)B &= 2\begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix} + (-1)\begin{bmatrix} 3 & -1 & 3 \\ -1 & 0 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 2 & 4 & 6 \\ 4 & 6 & 2 \end{bmatrix} + \begin{bmatrix} -3 & 1 & -3 \\ 1 & 0 & -2 \end{bmatrix} \\ &= \begin{bmatrix} -1 & 5 & 3 \\ 5 & 6 & 0 \end{bmatrix} \end{aligned}$$

Properties of matrix addition

The addition of matrices satisfy the following properties:

(i) **Commutative Law** If $A = [a_{ij}]$, $B = [b_{ij}]$ are matrices of the same order, say $m \times n$, then $A + B = B + A$.

Example 9. *Let* $A = \begin{bmatrix} 3 & 7 \\ 2 & 4 \end{bmatrix}$ *and* $B = \begin{bmatrix} 5 & 2 \\ 8 & 1 \end{bmatrix}$ *be two matrix. Then show that* $A + B = B + A$.

Solution:

$$A + B = \begin{bmatrix} 3 & 7 \\ 2 & 4 \end{bmatrix} + \begin{bmatrix} 5 & 2 \\ 8 & 1 \end{bmatrix} = \begin{bmatrix} 3+5 & 7+2 \\ 2+8 & 4+1 \end{bmatrix} = \begin{bmatrix} 8 & 9 \\ 10 & 5 \end{bmatrix} \tag{0.1}$$

$$B + A = \begin{bmatrix} 5 & 2 \\ 8 & 1 \end{bmatrix} + \begin{bmatrix} 3 & 7 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} 5+3 & 2+7 \\ 8+2 & 1+4 \end{bmatrix} = \begin{bmatrix} 8 & 9 \\ 10 & 5 \end{bmatrix} \tag{0.2}$$

Clearly, from (0.1) and (0.2) $A + B = B + A$.

(ii) **Associative Law** For any three matrices $A = [a_{ij}]$, $B = [b_{ij}]$, $C = [c_{ij}]$ of the same order, say $m \times n$, (A + B) + C = A + (B + C).

Example 10. *Consider* $A = \begin{bmatrix} 5 & 2 \\ 2 & 8 \end{bmatrix}$, $B = \begin{bmatrix} 3 & 4 \\ 9 & 6 \end{bmatrix}$ *and* $C = \begin{bmatrix} 1 & 0 \\ 3 & 7 \end{bmatrix}$. *Then show that* $(A + B) + C = A + (B + C)$.

Solution:

$$(A + B) + C = \left(\begin{bmatrix} 5 & 2 \\ 2 & 8 \end{bmatrix} + \begin{bmatrix} 3 & 4 \\ 9 & 6 \end{bmatrix} \right) + \begin{bmatrix} 1 & 0 \\ 3 & 7 \end{bmatrix} \tag{0.3}$$

$$= \left(\begin{bmatrix} 5+3 & 2+4 \\ 2+9 & 8+6 \end{bmatrix} \right) + \begin{bmatrix} 1 & 0 \\ 3 & 7 \end{bmatrix} \tag{0.4}$$

$$= \begin{bmatrix} 8 & 6 \\ 11 & 14 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 3 & 7 \end{bmatrix} = \begin{bmatrix} 8+1 & 6+0 \\ 11+3 & 14+7 \end{bmatrix} = \begin{bmatrix} 9 & 6 \\ 14 & 21 \end{bmatrix} \tag{0.5}$$

$$A + (B + C) = \begin{bmatrix} 5 & 2 \\ 2 & 8 \end{bmatrix} + \left(\begin{bmatrix} 3 & 4 \\ 9 & 6 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 3 & 7 \end{bmatrix} \right) \tag{0.6}$$

$$= \begin{bmatrix} 5 & 2 \\ 2 & 8 \end{bmatrix} + \left(\begin{bmatrix} 3+1 & 4+0 \\ 9+3 & 6+7 \end{bmatrix} \right) \tag{0.7}$$

$$= \begin{bmatrix} 5 & 2 \\ 2 & 8 \end{bmatrix} + \begin{bmatrix} 4 & 4 \\ 12 & 13 \end{bmatrix} = \begin{bmatrix} 5+4 & 2+4 \\ 2+12 & 8+13 \end{bmatrix} = \begin{bmatrix} 9 & 6 \\ 14 & 21 \end{bmatrix} \tag{0.8}$$

Hence, from (0.5) and (0.8) $(A + B) + C = A + (B + C)$

(iii) **Existence of additive identity** Let $A = [a_{ij}]$ be an $m \times n$ matrix and O be an $m \times n$ zero matrix, then $A + O = O + A = A$. In other words, O is the additive identity for matrix addition.

Example 11. *Consider matrix* $A = \begin{bmatrix} 3 & -1 \\ 7 & 9 \end{bmatrix}$ *and a zero matrix* $O = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$. *Then show that* $A + O = O + A = A$

Solution:

$$A + O = \begin{bmatrix} 3 & -1 \\ 7 & 9 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 3+0 & -1+0 \\ 7+0 & 9+0 \end{bmatrix} = \begin{bmatrix} 3 & -1 \\ 7 & 9 \end{bmatrix} = A \tag{0.9}$$

$$O + A = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 3 & -1 \\ 7 & 9 \end{bmatrix} = \begin{bmatrix} 0+3 & 0-1 \\ 0+7 & 0+9 \end{bmatrix} = \begin{bmatrix} 3 & -1 \\ 7 & 9 \end{bmatrix} = A \tag{0.10}$$

(iv) **The existence of additive inverse** Let $A = [a_{ij}]_{m \times n}$ be any matrix, then we have another matrix as $-A = [-a_{ij}]_{m \times n}$ such that $A + (-A) = (-A) + A = O$. So $-A$ is the additive inverse of A or negative of A.

Example 12. *Consider matrix* $A = \begin{bmatrix} -2 & 8 \\ -3 & 1 \end{bmatrix}$, *then* $-A = \begin{bmatrix} 2 & -8 \\ 3 & -1 \end{bmatrix}$. *Show that* $A + (-A) = (-A) + A = O$

Solution:

$$A + (-A) = \begin{bmatrix} -2 & 8 \\ -3 & 1 \end{bmatrix} + \begin{bmatrix} 2 & -8 \\ 3 & -1 \end{bmatrix} = \begin{bmatrix} -2+2 & 8-8 \\ -3+3 & 1-1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = O \tag{0.11}$$

$$A + (-A) = \begin{bmatrix} 2 & -8 \\ 3 & -1 \end{bmatrix} + \begin{bmatrix} -2 & 8 \\ -3 & 1 \end{bmatrix} = \begin{bmatrix} 2-2 & -8+8 \\ 3-3 & -1+1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = O \tag{0.12}$$

Properties of scalar multiplication of a matrix

If $A = [a_{ij}]$ and $B = [b_{ij}]$ be two matrices of the same order, say $m \times n$, and k and l are scalars, then

(i) $k(A + B) = kA + kB$,

Example 13. *Consider k=2 and two matrices* $A = \begin{bmatrix} 5 & 2 \\ 3 & 1 \end{bmatrix}$ *and* $B = \begin{bmatrix} 3 & 4 \\ 2 & 6 \end{bmatrix}$. *Show that* $k(A + B) = kA + kB$.

Solution:

$$k(A+B) = 2\left(\begin{bmatrix} 5 & 2 \\ 3 & 1 \end{bmatrix} + \begin{bmatrix} 3 & 4 \\ 2 & 6 \end{bmatrix}\right) = 2\left(\begin{bmatrix} 5+3 & 2+4 \\ 3+2 & 1+6 \end{bmatrix}\right) \tag{0.13}$$

$$= 2\begin{bmatrix} 8 & 6 \\ 5 & 7 \end{bmatrix} = \begin{bmatrix} 2\times 8 & 2\times 6 \\ 2\times 5 & 2\times 7 \end{bmatrix} = \begin{bmatrix} 16 & 12 \\ 10 & 14 \end{bmatrix} \tag{0.14}$$

$$kA + kB = 2\begin{bmatrix} 5 & 2 \\ 3 & 1 \end{bmatrix} + 2\begin{bmatrix} 3 & 4 \\ 2 & 6 \end{bmatrix} = \left(\begin{bmatrix} 10 & 4 \\ 6 & 2 \end{bmatrix} + \begin{bmatrix} 6 & 8 \\ 4 & 12 \end{bmatrix}\right. \tag{0.15}$$

$$= \begin{bmatrix} 10+6 & 4+8 \\ 6+4 & 2+12 \end{bmatrix} = \begin{bmatrix} 16 & 12 \\ 10 & 14 \end{bmatrix} \tag{0.16}$$

from (0.14) and (0.16) $k(A + B) = kA + kB$

(ii) $(k + l)A = kA + lA$.

Example 14. *Let* $k = 2$, $l = 3$ *and* $A = \begin{bmatrix} 6 & 9 \\ 7 & 4 \end{bmatrix}$, *then show* $(k + l)A = kA + lA$

Solution:

$$(k+l)A = \left(2+3\right)\begin{bmatrix} 6 & 9 \\ 7 & 4 \end{bmatrix} = 5\begin{bmatrix} 6 & 9 \\ 7 & 4 \end{bmatrix} = \begin{bmatrix} 30 & 45 \\ 35 & 20 \end{bmatrix} \tag{0.17}$$

$$kA + lA = 2\begin{bmatrix} 6 & 9 \\ 7 & 4 \end{bmatrix} + 3\begin{bmatrix} 6 & 9 \\ 7 & 4 \end{bmatrix} = \begin{bmatrix} 12 & 18 \\ 14 & 8 \end{bmatrix} + \begin{bmatrix} 18 & 27 \\ 21 & 12 \end{bmatrix} \tag{0.18}$$

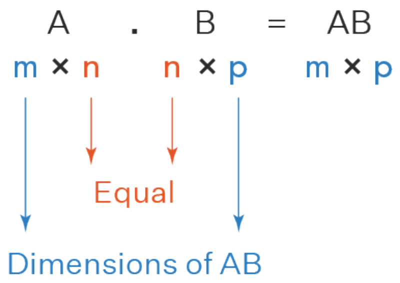
$$= \begin{bmatrix} 12+18 & 18+27 \\ 14+21 & 8+12 \end{bmatrix} = \begin{bmatrix} 30 & 45 \\ 35 & 20 \end{bmatrix} \tag{0.19}$$

Hence, from (0.17) and (0.19) $(k + l)A = kA + lA$.

Multiplication of matrices

The product of two matrices A and B is defined if the number of columns of A is equal to the number of rows of B. Let $A = [a_{ij}]$ be an $m \times n$ matrix and $B = [b_{jk}]$ be an $n \times p$ matrix.

Rule For Matrix Multiplication

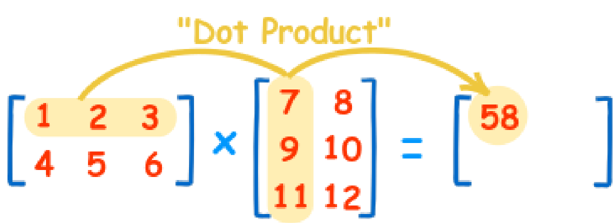


Example 15. *If* $C = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ *and* $C = \begin{bmatrix} 7 & 8 \\ 9 & 10 \\ 11 & 12 \end{bmatrix}$*, then find the product CD.*

Solution:

$$\mathbf{CD} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{23} \end{bmatrix}$$

To work out the answer for the **1st row** and **1st column** (a_{11}):

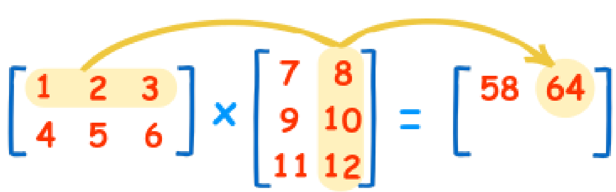


The "Dot Product" is where we multiply matching members, then sum up:

$$\mathbf{(1, 2, 3) \cdot (7, 9, 11) = 1 \times 7 + 2 \times 9 + 3 \times 11 = 58}$$

We match the 1st members (1 and 7), multiply them, likewise for the 2nd members (2 and 9) and the 3rd members (3 and 11), and finally sum them up.

Here it is for the **1st row** and **2nd column** (a_{12}):



$$(1, 2, 3) \cdot (8, 10, 12) = 1 \times 8 + 2 \times 10 + 3 \times 12 = 64$$

We can do the same thing for the **2nd row** and **1st column** (a_{21}):

$$(4, 5, 6) \cdot (7, 9, 11) = 4 \times 7 + 5 \times 9 + 6 \times 11 = 139$$

And for the **2nd row** and **2nd column** (a_{22}):

$$(4, 5, 6) \cdot (8, 10, 12) = 4 \times 8 + 5 \times 10 + 6 \times 12 = 154$$

And we get:

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \times \begin{bmatrix} 7 & 8 \\ 9 & 10 \\ 11 & 12 \end{bmatrix} = \begin{bmatrix} 58 & 64 \\ 139 & 154 \end{bmatrix}$$

Example 16. *Find AB, if* $A = \begin{bmatrix} 6 & 9 \\ 2 & 3 \end{bmatrix}$ *and* $B = \begin{bmatrix} 2 & 6 & 0 \\ 7 & 9 & 8 \end{bmatrix}$

Solution: The matrix A has 2 columns which is equal to the number of rows of B. Hence AB is defined. Now

$$\begin{aligned} AB &= \begin{bmatrix} 6 \times 2 + 9 \times 7 & 6 \times 6 + 9 \times 9 & 6 \times 0 + 9 \times 8 \\ 2 \times 2 + 3 \times 7 & 2 \times 6 + 3 \times 9 & 2 \times 0 + 3 \times 8 \end{bmatrix} \\ &= \begin{bmatrix} 12 + 63 & 36 + 81 & 0 + 72 \\ 4 + 21 & 12 + 27 & 0 + 24 \end{bmatrix} = \begin{bmatrix} 75 & 117 & 72 \\ 25 & 39 & 24 \end{bmatrix} \end{aligned}$$

Note: If AB is defined, then BA need not be defined. In the above example, AB is defined but BA is not defined because B has 3 column while A has only 2 (and not 3) rows.

Example 17. *If* $A = \begin{bmatrix} 1 & -2 & 3 \\ -4 & 2 & 5 \end{bmatrix}$ *and* $B = \begin{bmatrix} 2 & 3 \\ 4 & 5 \\ 2 & 1 \end{bmatrix}$*, , then find AB, BA. Show that* $AB \neq BA$.

Solution: Since A is a 2×3 matrix and B is 3×2 matrix. Hence AB and BA are both defined and are matrices of order 2×2 and 3×3, respectively. Note that

$$AB = \begin{bmatrix} 1 & -2 & 3 \\ -4 & 2 & 5 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ 4 & 5 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 2 - 8 + 6 & 3 - 10 + 3 \\ -8 + 8 + 10 & -12 + 10 + 5 \end{bmatrix} = \begin{bmatrix} 0 & -4 \\ 10 & 3 \end{bmatrix}$$

$$BA = \begin{bmatrix} 2 & 3 \\ 4 & 5 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & -2 & 3 \\ -4 & 2 & 5 \end{bmatrix} = \begin{bmatrix} 2 - 12 & -4 + 6 & 6 + 15 \\ 4 - 20 & -8 + 10 & 12 + 25 \\ 2 - 4 & -4 + 2 & 6 + 5 \end{bmatrix} = \begin{bmatrix} -10 & 2 & 21 \\ -16 & 2 & 37 \\ -2 & -2 & 11 \end{bmatrix}$$

Clearly $AB \neq BA$.
Thus matrix multiplication is **not commutative**.

Note: Multiplication of diagonal matrices of same order will be commutative.

Example 18. *Consider diagonal matrix* $A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$, *and* $B = \begin{bmatrix} 3 & 0 \\ 0 & 4 \end{bmatrix}$, *then* $AB = BA = \begin{bmatrix} 3 & 0 \\ 0 & 8 \end{bmatrix}$

Example 19. *Find AB, if* $A = \begin{bmatrix} 0 & -1 \\ 0 & -2 \end{bmatrix}$, *and* $B = \begin{bmatrix} 3 & 5 \\ 0 & 0 \end{bmatrix}$

Solution: We have $AB = \begin{bmatrix} 0 & -1 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} 3 & 5 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

$\Longrightarrow$ **This is an example where two matrix A and B are not zero matrix but their product is a zero matrix.**

Properties of multiplication of matrices

The multiplication of matrices possesses the following properties

(i) **The associative law:** For any three matrices A, B and C. We have $(AB)C = A(BC)$, whenever both sides of the equality are defined.
Verfication: Let $A = \begin{bmatrix} 1 & 4 \\ 2 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 3 & 6 \\ 1 & 4 \end{bmatrix}$ and $C = \begin{bmatrix} 1 & 0 \\ 3 & 2 \end{bmatrix}$.
We can find (AB)C as follows:

$$(AB)C = \left(\begin{bmatrix} 1 & 4 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} 3 & 6 \\ 1 & 4 \end{bmatrix} \right) \begin{bmatrix} 1 & 0 \\ 3 & 2 \end{bmatrix} = \begin{bmatrix} 7 & 22 \\ 8 & 20 \end{bmatrix} \bigg) \begin{bmatrix} 1 & 0 \\ 3 & 2 \end{bmatrix} = \begin{bmatrix} 73 & 44 \\ 68 & 40 \end{bmatrix}$$

And, we can find A(BC) as follows:

$$(AB)C = \begin{bmatrix} 1 & 4 \\ 2 & 2 \end{bmatrix} \left(\begin{bmatrix} 3 & 6 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 3 & 2 \end{bmatrix} \right) = \begin{bmatrix} 1 & 4 \\ 2 & 2 \end{bmatrix} \bigg) \begin{bmatrix} 21 & 12 \\ 13 & 8 \end{bmatrix} = \begin{bmatrix} 73 & 44 \\ 68 & 40 \end{bmatrix}$$

Hence, $(AB)C = A(BC)$.

(ii) **The distributive law:** For three matrices A, B and C.

(a) A (B+C) = AB + AC, whenever both sides of equality are defined.
Verfication: Let $A = \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 3 & 1 \\ 0 & 1 \end{bmatrix}$ and $C = \begin{bmatrix} 1 & 4 \\ 3 & 2 \end{bmatrix}$.
$A(B + C) = \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} \left(\begin{bmatrix} 3 & 1 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 4 \\ 3 & 2 \end{bmatrix} \right) = \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} \left(\begin{bmatrix} 4 & 5 \\ 3 & 3 \end{bmatrix} \right) = \begin{bmatrix} 11 & 13 \\ 18 & 21 \end{bmatrix}$
$AB + AC = \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 1 & 4 \\ 3 & 2 \end{bmatrix} = \begin{bmatrix} 6 & 3 \\ 9 & 5 \end{bmatrix} + \begin{bmatrix} 5 & 10 \\ 9 & 16 \end{bmatrix} = \begin{bmatrix} 11 & 13 \\ 18 & 21 \end{bmatrix}$
Hence, $A(B + C) = AB + AC$.

(b) (A+B) C = AC + BC, whenever both sides of equality are defined.
Verification do it yourself!

(iii) **The existence of multiplicative identity:** For every square matrix A, there exist an identity matrix of same order such that IA = AI = A.

Verification: Let $A = \begin{bmatrix} 3 & 5 \\ 8 & 7 \end{bmatrix}$, then

$$AI = \begin{bmatrix} 3 & 5 \\ 8 & 7 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 \times 1 + 5 \times 0 & 3 \times 0 + 5 \times 1 \\ 8 \times 1 + 7 \times 0 & 8 \times 0 + 7 \times 1 \end{bmatrix} = \begin{bmatrix} 3 & 5 \\ 8 & 7 \end{bmatrix}$$

$$IA = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3 & 5 \\ 8 & 7 \end{bmatrix} = \begin{bmatrix} 1 \times 3 + 0 \times 8 & 1 \times 5 + 0 \times 7 \\ 0 \times 3 + 1 \times 8 & 0 \times 5 + 1 \times 7 \end{bmatrix} = \begin{bmatrix} 3 & 5 \\ 8 & 7 \end{bmatrix}$$ So, $AI = IA = A$.

Transpose of a Matrix

If $A = [a_{ij}]$ be an $m \times n$ matrix, then the matrix obtained by interchanging the rows and columns of A is called the transpose of A. Transpose of the matrix A is denoted by A' or (A^T).

Example 20. *If* $A = \begin{bmatrix} 3 & 5 \\ \sqrt{3} & 1 \\ 0 & -\frac{1}{5} \end{bmatrix}_{3\times 2}$ *then* $A' = \begin{bmatrix} 3 & \sqrt{3} & 0 \\ 5 & 1 & -\frac{1}{5} \end{bmatrix}_{2\times 3}$

Properties of transpose of the matrices

For any matrices A and B of suitable orders, we have

(i) $(A')' = A$

Verification: Let $A = \begin{bmatrix} 3 & 5 \\ \sqrt{3} & 1 \\ 0 & -\frac{1}{5} \end{bmatrix} \implies A' = \begin{bmatrix} 3 & \sqrt{3} & 0 \\ 5 & 1 & -\frac{1}{5} \end{bmatrix} \implies (A')' = \begin{bmatrix} 3 & 5 \\ \sqrt{3} & 1 \\ 0 & -\frac{1}{5} \end{bmatrix}$=A

Thus $(A')' = A$.

(ii) $(kA)' = kA'$ (where k is any constant)(Discussed in class.)

(iii) $(A + B)' = A' + B'$ (Discussed in class.)

(iv) $(AB)' = B'A'$ (Discussed in class.)